Name:_____

Reactions with Acids

1. Acid and Base. Neutralisation reactions. Here and acid and base are mixed together. This makes a salt and water.

Acid + Base ⟹ salt + water

Hydrochloric Acid + sodium Hydroxide ⟹ sodium chloride + water

HCl + NaOH ⟹ NaCl + H_2O

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2. Acid and Metal reaction. This makes Hydrogen gas and a salt.

Acid + Metal ⟹ Hydrogen gas + salt

Hydrochloric Acid + Magnesium ⟹ Magnesium Chloride + Hydrogen gas

Acid + Metal ⟹ gas + salt

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3. Acid and carbonate. Remember the carbonate ion is:

$$CO_3^{-2}$$

It is a radical ion. Any compound with CO_3^{-2} in it is called a carbonate. For example

Calcium carbonate= $CaCO_3$

An acid added to a carbonate makes carbon dioxide gas, water and a salt.

Acid + carbonate ⟹ Carbon dioxide + water + salt

Nitric Acid + sodium carbonate ⟹ carbon dioxide + water + sodium nitrate

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Complete this table

Equation	Type of reaction? Acid+base, acid+metal acid+carbonate
Nitric acid+sodium carbonate--->sodium nitrate+carbon	

dioxide+water	
Hydrochloric acid + zinc ---> zinc chloride + hydrogen gas	
$H_2SO_4 + 2KOH \rightarrow K_2SO_4 + 2H_2O$	
$2HCl + CaCO_3 \rightarrow CaCl_2 + CO_2 + H_2O$	
Nitric acid + calcium hydroxide ---> calcium nitrate + water	
$2HCl + Zn \rightarrow ZnCl_2 + H_2$	
$H_2SO_4 + Mg \rightarrow MgSO_4 + H_2$	

Complete these equations.

Acid + base ⟹ salt+_____

Acid + metal ⟹ salt +_____ ___

Acid + carbonate ⟹ salt+_____ _____ +____

_____+_____ ⟹ salt+water+carbon dioxide

_____+_____ ⟹ salt+hydrogen gas

_____+_____ ⟹ salt +water

Hydrochloric acid+magnesium metal ---> magnesium chloride+_____ gas

HCl + KOH ---> KCl + _____

HCl + $MgCO_3$ ---> $MgCl_2$ + _____ + _____

Name	Formula
Hydrochloric acid	

Sulfuric acid	
Nitric acid	
Acetic acid (vinegar)	
	H_2O
	CO_2
	CO_3^{2-}
Sodium Hydroxide	
Calcium Carbonate	